CAPSTONE PROJECT

AI SMS SCAM DETECTOR

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OUTLINE

- Problem Statement
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- System Development Approach
- Algorithm & Deployment
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PROBLEM STATEMENT

In India, SMS-based financial scams cause massive losses every year (₹3,600+ crore reported in 2025 alone – NCRB data).

Common scams include:

- *UPI KYC update fraud*
- *Digital arrest / police impersonation*
- *Lottery / prize refund scams*
- *Fake government scheme offers*

Most people (especially rural and low-literacy users) cannot identify these messages in time.

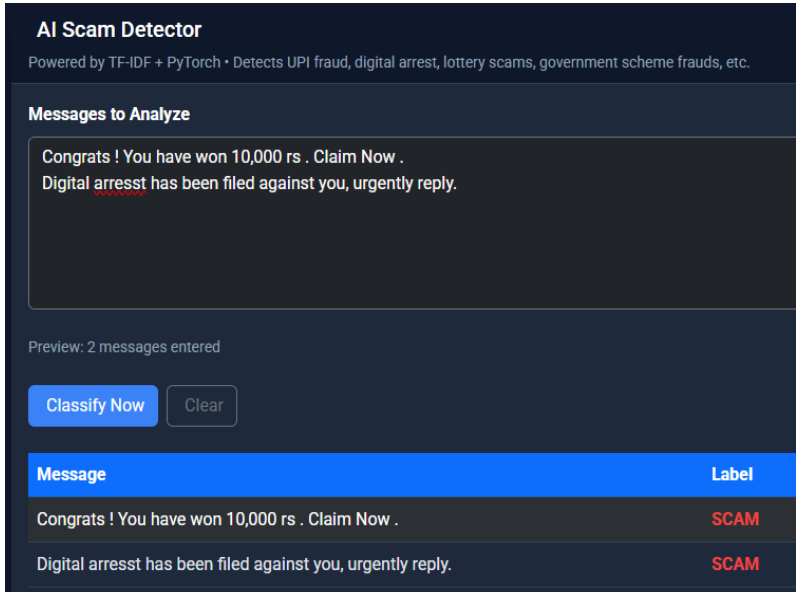
There is a need for a fast, accurate, and easy-to-use tool that can instantly classify suspicious SMS messages as **SCAM** or **LEGITIMATE**.

PROPOSED SOLUTION

ScamShield – an AI-powered, lightweight web application that:

- *Allows users to paste suspicious messages (single or batch)*
- *Instantly classifies them as SCAM or LEGITIMATE*
- *Shows confidence percentage*
- *Provides color-coded results (red = scam, green = legitimate)*
- *Saves history in log file for review*
- *Runs completely offline after training*
- *Works on low-spec laptops and mobile browsers*

Goal: Empower common users to detect scams before losing money.



SYSTEM APPROACH

System Development Approach & Technology Used -

- Programming Language: Python 3.12
- Web Framework: Flask (lightweight backend)
- UI Framework: Bootstrap 5 (responsive dark-mode design)
- Text Processing: TF-IDF Vectorization (scikit-learn)
- Machine Learning Model: Simple Feed-Forward Neural Network (PyTorch)
- Dataset: Custom balanced JSON dataset (scam vs legitimate messages)
- Logging: Auto-save results to scam_log.txt
- Additional Libraries: numpy, torch, rich (CLI), jsPDF (PDF export)

Zero-cost, open-source stack – runs on low-spec hardware.

ALGORITHM & DEPLOYMENT

Algorithm Flow:

1. User pastes messages (single or batch).
2. Text is preprocessed (lowercase, remove punctuation).
3. TF-IDF vectorization converts text to numerical features.
4. Pre-trained neural network predicts scam (1) or legitimate (0).
5. Softmax gives confidence score.
6. Results displayed with colors, logged to file.

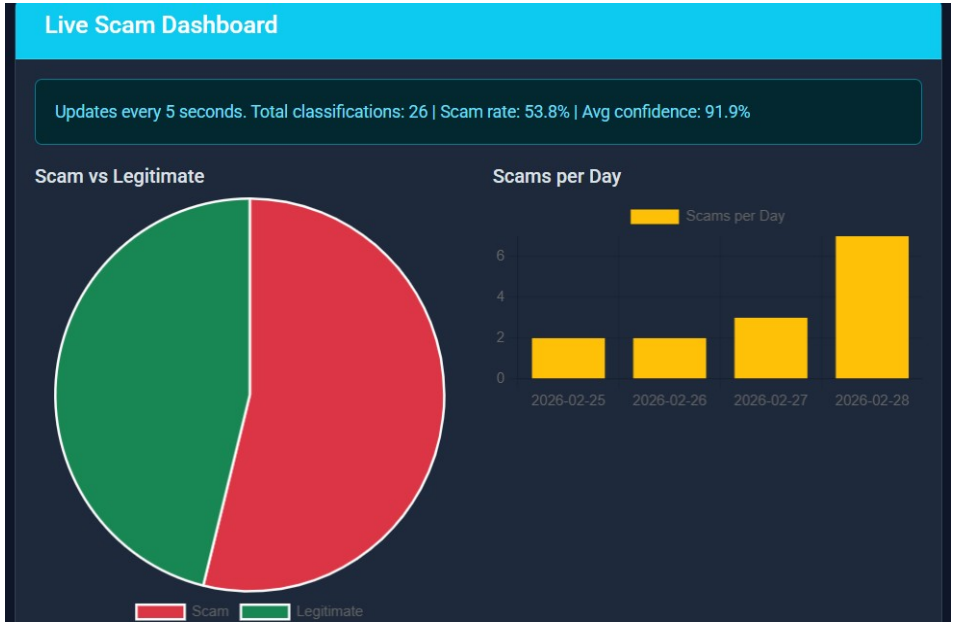
Deployment:

- *Local: Run `python main.py` → choose GUI or CLI as needed.*
- *Online: Deployed on free platform of Render.com.*
- *No external server or API needed – fully offline after training.*

RESULT

- ***Training:*** *50 epochs,* ***final loss:*** *0.0034*
- ***Test Accuracy****: 90.91% on unseen messages*

- ***Key Features Delivered:***
  - *Batch processing*
  - *Confidence scoring (with low-conf warning)*
  - *Real-time preview*
  - *Color-coded results*
  - *PDF export*
  - *Auto-logging*
  - *Responsive UI*



Test Case Examples:

- *Scam: "UPI KYC pending 24hr block click now" → SCAM (94%)*
- *Legitimate: "SBI: Rs 5000 credited. Ref 123456" → LEGITIMATE (96%)*

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CONCLUSION

ScamShield successfully achieves its goal of providing a simple, accurate, and accessible tool to detect common Indian scam messages.

Key Achievements:

- *High accuracy (90.91%) with lightweight model*
- *Zero-cost, low-resource implementation*
- *User-friendly GUI + CLI*
- *Real-world relevance for Indian users*

This project demonstrates practical application of machine learning and web app integration to solve a serious social problem.

FUTURE SCOPE

- **Mobile Application Development:**
  - *Android/iOS app with automatic SMS scanning & real-time scam alerts.*
  - *Offline model inference using TensorFlow Lite / PyTorch Mobile.*

- **Multilingual & Indic Language Expansion:**
  - *Use IndicBERT or MuRIL for code-mixed messages.*
  - *Retrain on multilingual scam dataset collected from Indian users.*

- **Privacy, Security & Compliance:**
  - *Full on-device processing (no data sent to server).*
  - *Compliance with Digital Personal Data Protection Act 2023 (DPDP Act).*
  - *End-to-end encryption for any user-submitted messages.*

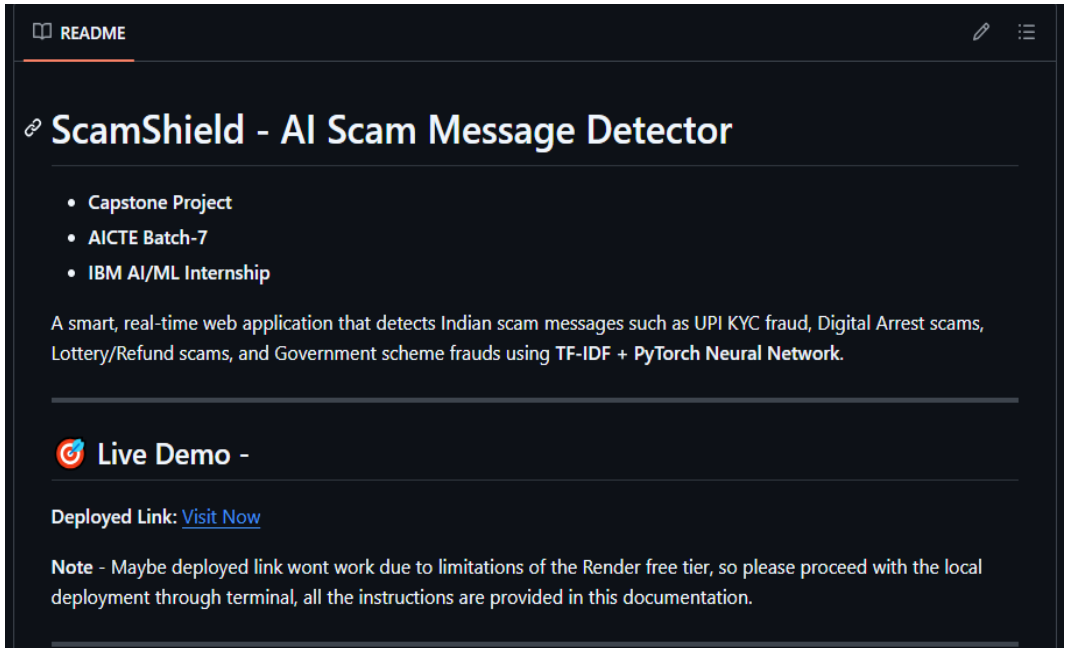
- **Advanced Model & Accuracy Improvements:**
  - *Upgrade to LSTM / BERT-tiny for better context understanding.*
  - *Add explainable AI (highlight which words triggered scam label).*
  - *WhatsApp/Telegram bot for forwarding suspicious messages.*
  - *One-click reporting to **Cyber helpline 1930** (pre-filled complaint).*

REFERENCES

1. **Supervised learning —** scikit-learn 1.8.0 documentation
2. **PyTorch official documentation –** pytorch.org
3. **Flask official documentation –** flask.palletsprojects.com
4. **Bootstrap 5 –** getbootstrap.com
5. **jsPDF library –** github.com/parallax/jsPDF
6. NCRB Crime in India Report 2025 (cyber fraud statistics)

GITHUB LINK

- **Github Link :** https://github.com/kushal-soni-cyber/Scam-Shield
- **Deployment Link:** https://scam-shield-7mhi.onrender.com/ (Render)



README

ScamShield - AI Scam Message Detector

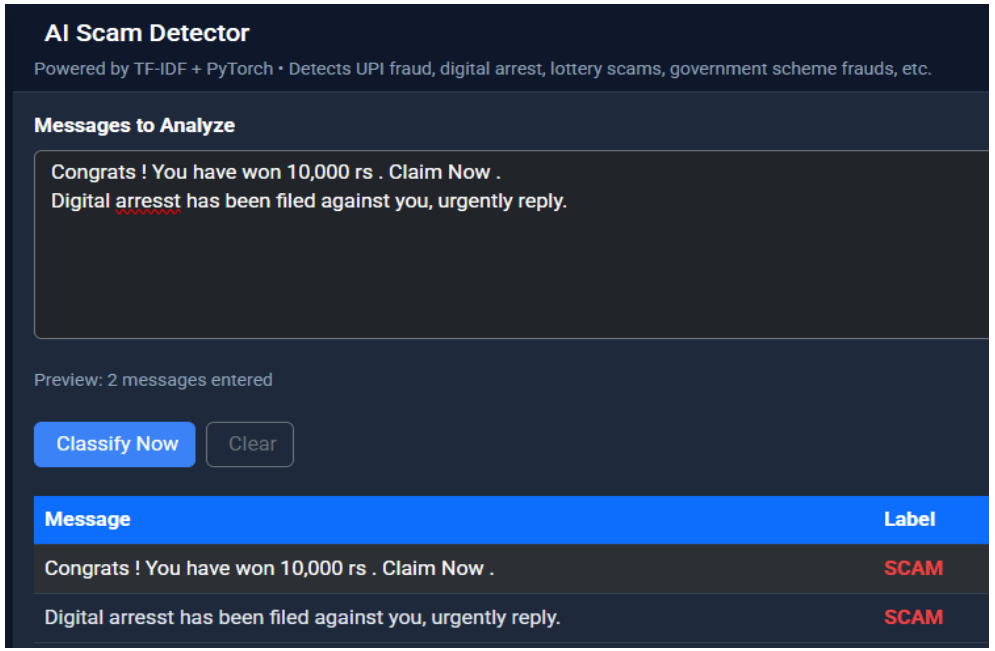
- Capstone Project
- AICTE Batch-7
- IBM AI/ML Internship

A smart, real-time web application that detects Indian scam messages such as UPI KYC fraud, Digital Arrest scams, Lottery/Refund scams, and Government scheme frauds using **TF-IDF + PyTorch Neural Network**.

🎯 Live Demo -

Deployed Link: Visit Now

Note - Maybe deployed link wont work due to limitations of the Render free tier, so please proceed with the local deployment through terminal, all the instructions are provided in this documentation.



AI Scam Detector

Powered by TF-IDF + PyTorch • Detects UPI fraud, digital arrest, lottery scams, government scheme frauds, etc.

Messages to Analyze

Congrats ! You have won 10,000 rs . Claim Now .
Digital arresst has been filed against you, urgently reply.

Preview: 2 messages entered

Classify Now | Clear

Message	Label
Congrats ! You have won 10,000 rs . Claim Now .	SCAM
Digital arresst has been filed against you, urgently reply.	SCAM

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THANK YOU

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